



# Python for Fixed Income

COURSE 2 · BOND MATH

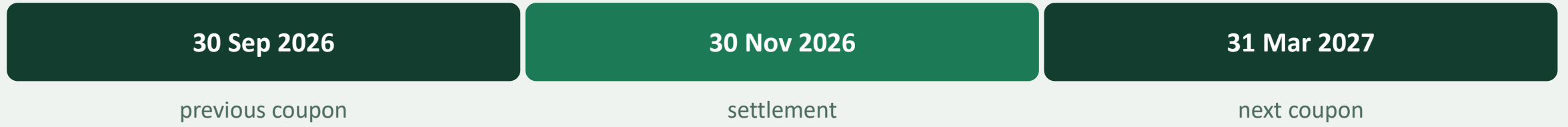
## Day Counts and Accrued Interest

WEEK 2 · DAY 2

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# Who earns the coupon



## The seller held it: 2 months

Earned that interest, but will not be paid it

## The buyer holds it: 4 months

Collects the whole coupon on 31 March

**At settlement the buyer pays the seller the accrued interest, on top of the price.**

**$\text{accrued} = \text{coupon\_pct} / \text{frequency} \times (\text{days accrued}) / (\text{days in the period})$**

How to count those days is set by the bond's documents, not by the calendar.

# Two counts, two answers

**0.879808**

calendar days: 61 of 182, per 100

**0.875000**

30/360: 60 of 180, per 100

**How many days is a convention, not a fact of the calendar.**

Each bond's documents name its day count. The course's bonds, like US corporate bonds generally, are 30/360.

**Quorvane, 5.25 percent, 30 Nov 2026**

One coupon **5.25 / 2 = 2.625**

Excel's accrued **0.875000**

On 1,000,000 face **8,750.00 dollars**

Calendar, more by **48.08 dollars**

# The four US 30/360 rules

$$\text{Days} = 360 \times (\text{Y2} - \text{Y1}) + 30 \times (\text{M2} - \text{M1}) + (\text{D2} - \text{D1})$$

| Rule | Change                                                |
|------|-------------------------------------------------------|
| 1    | Start is the last day of February: D1 = 30            |
| 2    | Start and end both last day of February: D2 = 30      |
| 3    | D1 is 31: D1 = 30                                     |
| 4    | D2 is 31 and the start's own day is 30 or 31: D2 = 30 |

| Case       | Dates                    | No rule | Rules | Excel     |
|------------|--------------------------|---------|-------|-----------|
| No rule    | 2026-08-16 to 2026-09-30 | 44      | 44    | COUPDAYBS |
| Rule 1     | 2027-02-28 to 2027-03-01 | 3       | 1     | COUPDAYBS |
| Rule 2     | 2027-02-28 to 2028-02-29 | 361     | 360   | YEARFRAC  |
| Rule 3     | 2026-08-31 to 2026-09-30 | 29      | 30    | COUPDAYBS |
| Rule 4     | 2026-07-30 to 2026-10-31 | 91      | 90    | COUPDAYBS |
| Not rule 4 | 2026-10-15 to 2026-10-31 | 16      | 16    | COUPDAYBS |
| Own day    | 2027-02-28 to 2027-03-31 | 33      | 31    | COUPDAYBS |

Every rule is about the end of a month. Rule 4 asks about the start date's own day.

# days\_30\_360, and a lookalike

```
# rule 1: the last day of February counts as the 30th
if start_is_feb_end: d1 = 30
# rule 2: both dates on the last day of February
if start_is_feb_end and end_is_feb_end: d2 = 30
# rule 3, then rule 4 on start.day, not d1
if start.day == 31: d1 = 30
if end.day == 31 and start.day >= 30: d2 = 30
```

The notebook's function writes each rule as an if block with its comment; the logic is the same.

## Excel side by side (tested in Excel)

|                                                   |     |
|---------------------------------------------------|-----|
| =YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360  | 360 |
| =DAYS360(DATE(2027,2,28),DATE(2028,2,29))         | 359 |
| =YEARFRAC(DATE(2026,2,28),DATE(2026,10,31),0)*360 | 241 |
| =DAYS360(DATE(2026,2,28),DATE(2026,10,31))        | 240 |
| =DAYS360(DATE(2027,2,28),DATE(2027,2,28))         | -2  |

**DAYS360 has its own February rules. The course uses the count COUPDAYBS and PRICE use.**

# Actual/actual: every count in calendar days

**61 / 182**

actual/actual: accrued 0.879808 per 100

**60 / 180**

30/360: accrued 0.875000 per 100

Days accrued are calendar days, and E is the calendar length of the coupon period: 182 or 183 days for a half year.

E under actual/actual is COUPNCD - COUPPCD, the length PRICE uses.

## Excel side by side (tested in Excel)

|                                                              |     |
|--------------------------------------------------------------|-----|
| <code>=COUPDAYBS(DATE(2026,9,30),DATE(2031,4,30),2,1)</code> | 153 |
| <code>=COUPDAYS(DATE(2026,9,30),DATE(2031,9,30),2,1)</code>  | 181 |

The second is the period from 30 September 2026 to 31 March 2027: 182 calendar days. COUPDAYS with basis 1 differs from the period's length in 12 of the 28 basis 1 rows.

# A, E, and DSC: coupon\_period\_days

| Basis   | A  | E   | DSC |
|---------|----|-----|-----|
| 30/360  | 60 | 180 | 120 |
| ACT/ACT | 61 | 182 | 121 |

A: previous coupon to settlement. E: days in the period. DSC: settlement to next coupon. Under 30/360, DSC = E - A, what PRICE uses.

## Excel side by side (tested in Excel)

|                                                   |     |
|---------------------------------------------------|-----|
| =COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)  | 60  |
| =COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)   | 180 |
| =COUPDAYSNC(DATE(2026,11,30),DATE(2031,9,30),2,0) | 120 |

## Bond 99031FAF8, 30 September 2026

Previous coupon 2026-08-28

Next coupon 2027-02-28, a February month end

|            |            |                       |
|------------|------------|-----------------------|
| E - A      | <b>148</b> | what PRICE uses       |
| COUPDAYSNC | <b>150</b> | not used under 30/360 |
| Calendar   | <b>151</b> | actual days           |

Three answers to days to the next coupon. The course follows PRICE.

# accrued\_interest, and ACCRINT

| Issuer          | Coupon | Accrued per 100 |
|-----------------|--------|-----------------|
| Quorvane Energy | 5.25   | 0.875000        |
| Dunmarrow Rail  | 6.125  | 1.020833        |
| Pellwick Paper  | 4.75   | 0.791667        |
| Halvering Foods | 7.5    | 1.250000        |

`return coupon_pct / frequency * days_accrued / days_in_period`

## Where ACCRINT differs: 99031FAF8

Next coupon 28 February 2027 is a month end, so ACCRINT counts from 31 August: 33 days, 0.7448. The bond's own schedule: 32 days, 0.7222.

## Excel side by side (tested in Excel)

|                                                                                |       |
|--------------------------------------------------------------------------------|-------|
| <code>=100*5.25/100/2*COUPDAYBS(D)/COUPDAYS(D)</code>                          | 0.875 |
| <code>=ACCRINT(COUPPCD(D),COUPNCD(D),DATE(2026,11,30),5.25/100,100,2,0)</code> | 0.875 |

D stands for the four COUP arguments: DATE(2026,11,30),DATE(2031,9,30),2,0

**The course follows the COUPDAYBS formula, the way PRICE counts. ACCRINT is a second check.**

# The holdings accrued column, reproduced

**200**

bonds, every one within rounding of the file's 3 decimals

**0.0005000000**

largest gap per 100: half of the third decimal

**7**

bonds at a rounding tie, such as 0.0875 filed as 0.087

**So the check allows  $0.0005 + 1e-9$  on the unrounded value: a hair of floating point sits on top of every tie.**

99031FAF8 is in the file at 0.722: the COUPDAYBS route, not ACCRINT's 0.7448.

Against Excel's unrounded values in the reference file, all 200 agree within the test's  $1e-9$ .

# Today's Excel, every formula tested

| Job               | Excel (tested), Quorvane on 30 Nov 2026                  | Returns |
|-------------------|----------------------------------------------------------|---------|
| Days accrued, A   | =COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)         | 60      |
| Days in period, E | =COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)          | 180     |
| Days to next      | =COUPDAYSNCD(DATE(2026,11,30),DATE(2031,9,30),2,0)       | 120     |
| Accrued per 100   | =100*5.25/100/2*COUPDAYBS(...)/COUPDAYS(...)             | 0.875   |
| Second route      | =ACCRINT(COUPPCD(...),COUPNCD(...),...,5.25/100,100,2,0) | 0.875   |
| 30/360 days       | =YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360         | 360     |
| A lookalike       | =DAYS360(DATE(2027,2,28),DATE(2028,2,29))                | 359     |

The /100 is in every formula: Excel takes the coupon as a decimal, bondmath in percent.

# Review a suggestion by its word diff

An assistant "tidies" coupon\_period\_days: a year has 365 days, so use 365 // frequency. It runs.

**git diff bondmath.py**

```
-     days_in_period = 360 // frequency
+     # a year has 365 days
+     days_in_period = 365 // frequency
```

**git diff --word-diff bondmath.py**

```
{+# a year has 365 days+}
days_in_period = [-360-]{+365+} // frequency
```

The comment is true of the calendar and beside the point: the bond is 30/360. The tests fail on the first reference row.

**Read the word diff, run the tests, then git restore bondmath.py: the committed file is back.**

# Bring it home: paste, test, read, commit

1

## Your folder

Set-Location to my\_bondmath, in your projects folder.

2

## Paste

Today's three functions and three tests at the end of your files, without the %%writefile line.

3

## Test

python -m pytest -q: 18 passed, or 24 with the exercises of days 1 to 6.

4

## Read, commit

git diff --word-diff bondmath.py, then git add and git commit.

5

## The habit

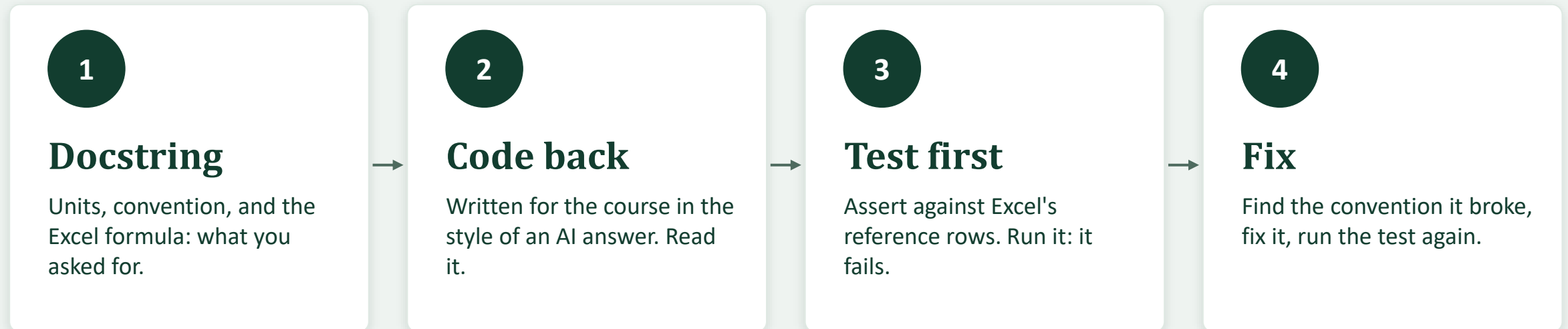
Any suggested edit: word diff, tests, and git restore if it is wrong.

```
git diff --word-diff bondmath.py
```

In Colab: download bondmath.py and test\_bondmath.py from the work folder under /tmp. They replace yesterday's downloads.

# The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.



**Today: accrued counts actual days over 365 for a 30/360 bond.**

*Code that runs is not the same as code that is right.*

# What you can do now

- Say why the buyer pays accrued interest: one coupon times  $A / E$
- Apply the four US 30/360 rules, and say why DAYS360 differs
- Count a coupon period on actual/actual
- Return A, E, and DSC, and accrued, matched to Excel on every row
- Review a suggested change with `git diff --word-diff`, and restore it

**Week 2, Day 3** Clean and dirty price: the quote, the invoice, and the price between coupons